

Part 1 — Carbon Fiber Arm Tube

DD_ Variable	Parameter	Source
DD_ArmLengthMm	Tube length (arm reach)	Computed: propeller radius + clearance margin + body radius
DD_ArmOdMm	Tube outer diameter	Motor max thrust → ComputeArmTubeOdMm() (10 / 12 / 16 / 20 / 25 mm stock)
DD_ArmIdMm	Tube inner diameter	Depends on the available standard carbon tube stock
Parameter keys injected by MacroRunner (read these in the macro):		
ARM_LENGTH_MM		
ARM_OD_MM		
ARM_ID_MM		
Arm length formula: ArmLengthMm = (PropDiameterMm / 2) + clearance_margin + body_radius Propeller diameter comes from PropellerSpec.DiameterInches × 25.4.		

Part 2 — Body-to-Arm Connection

DD_ Variable	Parameter	Source
DD_ArmOdMm	Arm tube outer diameter (bore to grip)	Same as arm tube DD_ArmOdMm
DD_ClampLengthMm	Clamp length along the arm	Design choice (typically 1.5 × arm OD)
DD_ClampBoltMm	Clamping bolt diameter	Design choice (M3 = 3.0, M4 = 4.0)
DD_ClampBoltSpacingMm	Spacing between clamp bolts	Derived from arm OD
DD_ChassisAttachSpacingMm	Bolt spacing for chassis attachment	Must match chassis plate hole pattern
DD_ChassisAttachBoltMm	Chassis attachment bolt diameter	Design choice (M3 = 3.0)
DD_WallThicknessMm	Part wall / material thickness	Design choice (3–5 mm for carbon or aluminium)
DD_ChassisSpacing	Distance betweent the chassis plates	Design choice (50 mm for the tallest Raspberry pi module + 10 mm clearance)
Parameter keys injected by MacroRunner:		
ARM_OD_MM		
CLAMP_LENGTH_MM		
CLAMP_BOLT_MM		
CLAMP_BOLT_SPACING_MM		
CHASSIS_ATTACH_SPACING_MM		
CHASSIS_ATTACH_BOLT_MM		
WALL_THICKNESS_MM		
CHASSIS_SPACING		

Part 3 — Chassis Plates

The chassis receives parameters from several components at once. Group them in the macro by source for clarity.		
From flight controller spec		
DD_ Variable	Parameter	Source
DD_FcMountPatternMm	Mounting hole pattern (square side length)	FlightControllerSpec — standard values: 20, 25.5, 30.5 mm
DD_FcLengthMm	FC board length	FlightControllerSpec.Dimensions.Length
DD_FcWidthMm	FC board width	FlightControllerSpec.Dimensions.Width
DD_FcHoleMm	FC mounting hole diameter	M3 clearance = 3.2 mm
From onboard computer spec		
DD_ Variable	Parameter	Source
DD_OcLengthMm	Board length	OnboardComputerSpec.Dimensions.Length
DD_OcWidthMm	Board width	OnboardComputerSpec.Dimensions.Width
DD_OcMountPatternMm	Mounting hole pattern	OnboardComputerSpec (varies by board)
DD_OcHoleMm	Mounting hole diameter	M2.5 = 2.7 mm or M3 = 3.2 mm
From GPS module spec		
DD_ Variable	Parameter	Source
DD_GpsDiamMm	Module diameter (if circular)	GpsSpec.Dimensions
DD_GpsLengthMm	Module length (if rectangular)	GpsSpec.Dimensions.Length
DD_GpsWidthMm	Module width	GpsSpec.Dimensions.Width
DD_GpsHoleMm	Mount hole diameter	M2 = 2.2 mm or M3 = 3.2 mm
From battery spec		
DD_ Variable	Parameter	Source
DD_BattLengthMm	Battery length	BatterySpec.Dimensions.Length
DD_BattWidthMm	Battery width	BatterySpec.Dimensions.Width
DD_BattHeightMm	Battery height	BatterySpec.Dimensions.Height
From frame / mission config		
DD_ Variable	Parameter	Source
DD_ArmCount	Number of arms	MissionSpecs.Configuration
DD_ChassisRadiusMm	Chassis outer radius	Computed from arm count and attachment geometry
DD_PlateThicknessMm	Plate thickness	Design choice (2–4 mm carbon fibre)
DD_ArmAttachSpacingMm	Arm attachment bolt spacing	Must match body-to-arm connection
Parameter keys injected by MacroRunner:		
FC_MOUNT_PATTERN_MM		
FC_LENGTH_MM		
FC_WIDTH_MM		
FC_HOLE_MM		
OC_LENGTH_MM		
OC_WIDTH_MM		
OC_MOUNT_PATTERN_MM		
OC_HOLE_MM		
GPS_DIAM_MM		
GPS_LENGTH_MM		
GPS_WIDTH_MM		

GPS_HOLE_MM		
BATT_LENGTH_MM		
BATT_WIDTH_MM		
BATT_HEIGHT_MM		
ARM_COUNT		
CHASSIS_RADIUS_MM		
PLATE_THICKNESS_MM		
ARM_ATTACH_SPACING_MM		

Part 4 — Landing Gear Connection Part

DD_ Variable	Parameter	Source
DD_LgTubeOdMm	Landing gear tube outer diameter (bore to grip)	Computed from total drone weight
DD_LgTubIdMm	Landing gear tube inner diameter	Design choice / standard tube stock
DD_LgAngleDeg	Attachment angle relative to chassis	Design choice (0° vertical or 10–15° outward splay)
DD_LgMountSpacingMm	Chassis attachment bolt spacing	Must match chassis plate hole pattern
DD_LgMountBoltMm	Chassis attachment bolt diameter	Design choice (M3 = 3.0)
DD_WallThicknessMm	Part wall / material thickness	Design choice
Parameter keys injected by MacroRunner:		
LG_TUBE_OD_MM		
LG_TUBE_ID_MM		
LG_ANGLE_DEG		
LG_MOUNT_SPACING_MM		
LG_MOUNT_BOLT_MM		
WALL_THICKNESS_MM		

Part 5 — Landing Gear Tubes

DD_ Variable	Parameter	Source
DD_LgLengthMm	Tube length (leg height)	Computed: battery height + prop ground-tilt clearance + margin
DD_LgOdMm	Tube outer diameter	Same value as DD_LgTubeOdMm in connection part
DD_LgIdMm	Tube inner diameter	Design choice / standard tube stock
Parameter keys injected by MacroRunner:		
LG_LENGTH_MM		
LG_OD_MM		
LG_ID_MM		
Minimum leg length formula:		
LgLengthMm = BattHeightMm + (PropDiameterMm × sin(MaxTiltAngleDeg)) + ground_clearance_margin		
MaxTiltAngleDeg is typically 15°. ground_clearance_margin is typically 20–30 mm.		